

Assignment1

2024-09-11

Basic SIR model

this function sets up the basic “closed” SIR model

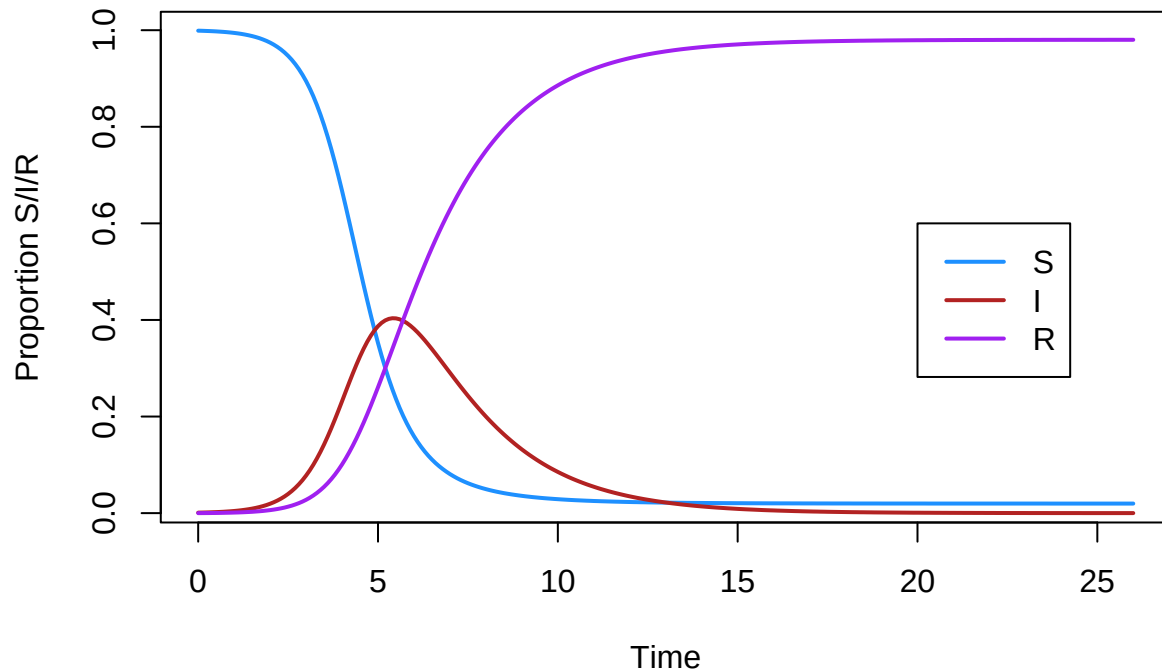
```
library(deSolve)

sirmod = function(t, startconds, parms) { # inputs into the model are "t"
  #- time, startconds = starting conditions and parms = parameters
  S = startconds[1]
  I = startconds[2]
  R = startconds[3]
  beta = parms["beta"]
  gamma = parms["gamma"]
  N = parms["N"]
  dS = - beta * S * I/N
  dI = beta * S * I/N - gamma * I
  dR = gamma * I
  res = c(dS, dI, dR)
  list(res)
}

## inputs into the model
times = seq(0, 26, by = 1/10) # weeks
parms = c(N = 1, beta = 2, gamma = 1/2) # in weeks
startconds = c(S = 0.999, I = 0.001, R = 0)

# run the model
out = ode(y = startconds, times = times, func = sirmod, parms = parms)
out = as.data.frame(out)

plot(times, out$S,type= "l", lwd = 2, col="dodgerblue",xlab = "", ylab = "")
title(xlab = "Time")
title(ylab = "Proportion S/I/R")
lines(times, out$I, lwd = 2, col = "firebrick")
lines(times, out$R, lwd = 2, col = "purple")
legend(20,0.6,lty = c(1,1,1), lwd = c(2,2,2),col = c("dodgerblue", "firebrick","purple"), legend = c("S", "I", "R"))
```



1a) what is the epidemic peak size (proportion infected at the peak of the outbreak)?

```
N <- parms['N']
I <- out['I']
max(I)/max(N)
```

```
## [1] 0.4035989
```

The epidemic peak size is .404.

1b) what is the total proportion of the population ever infected?

```
library('dplyr')
```

```
##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##   filter, lag
##
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library('tidyr')
R_last <- out %>%
  pull('R') %>%
  last()

I_last <- out %>%
  pull('I') %>%
  last()
```

```
R_last + I_last
```

```
## [1] 0.9801893
```

98.01% of the population is ever infected. Nobody passes back from recovered to infected or susceptible, so the total proportion of the population that is ever infected is the recovered portion at the end of the pandemic, plus the remaining infected (which is basically 0).

1c) what is the infectious period (/duration of infectiousness) for this model?

$D = 2$. (D is the inverse of γ and γ is set to $1/2$)

1d) what is the R_0 of this infection?

```
R0 = beta/gamma
```

```
R0 <- 2/(1/2)
```

```
R0
```

```
## [1] 4
```

```
R0 = 4
```

1e) Plot $R_{\text{effective}}$. In which week does $R_{\text{effective}}$ go below 1? What happens to cases at this point?

```
R_eff = R0 * S.
```

```
library(ggplot2)
```

```
library(latex2exp)
```

```
out$R_eff <- out$S * R0
```

```
ggplot(out, aes(x = time, y = R_eff)) +
```

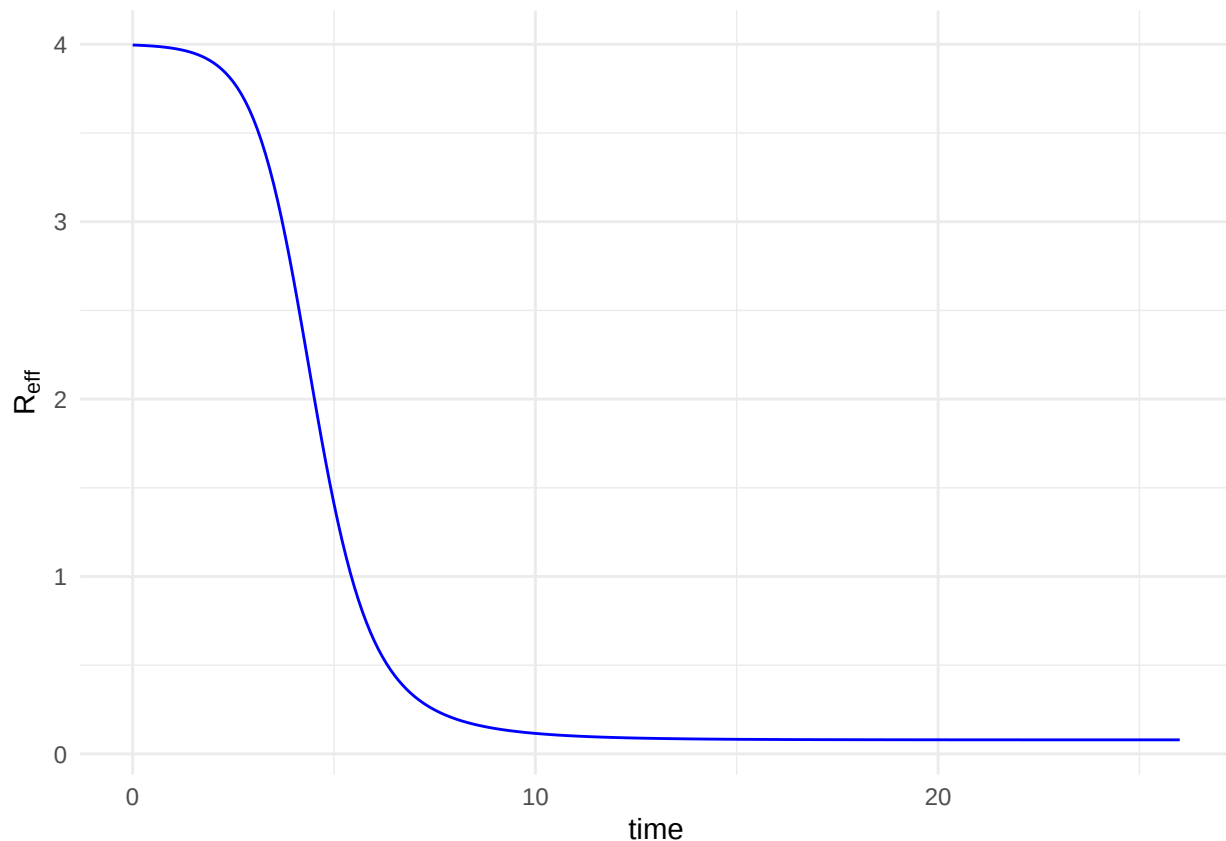
```
  geom_line(color = 'blue') +
```

```
  labs(
```

```
    y = TeX('$R_{\text{eff}}$')
```

```
  ) +
```

```
  theme_minimal()
```



```
R_eff_less <- out %>%
  filter(R_eff < 1) %>%
  slice(1) %>%
  pull(time)
R_eff_less
```

```
## [1] 5.5
```

R_e dips below 1 in the 6th week. Cases begin to decrease at this point, because the rate of recovery (γ) exceeds the rate of new infections (β).

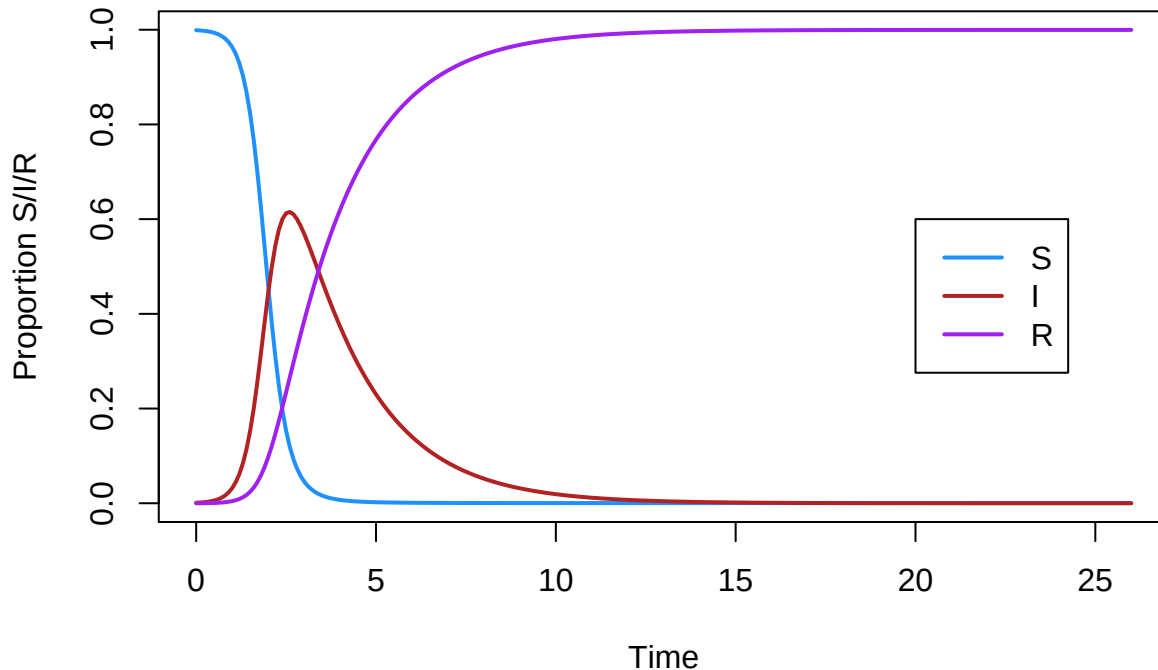
1f) Re-run the model assuming R_0 has doubled (assume the infectious period stays the same). Plot the output with overlaid S, I and R time series.

```
## inputs into the model
parms2 = c(N = 1, beta = 4, gamma = 1/2) # in weeks
startconds = c(S = 0.999, I = 0.001, R = 0)

# run the model
out2 = ode(y = startconds, times = times, func = sirmod, parms = parms2)
out2 = as.data.frame(out2)

plot(times, out2$S, type = "l", lwd = 2, col = "dodgerblue", xlab = "", ylab = "")
title(xlab = "Time")
title(ylab = "Proportion S/I/R")
lines(times, out2$I, lwd = 2, col = "firebrick")
lines(times, out2$R, lwd = 2, col = "purple")
```

```
legend(20,0.6,lty = c(1,1,1), lwd = c(2,2,2),col = c("dodgerblue", "firebrick","purple"), legend = c("S", "I", "R"))
```



##

What is the proportion infected at the peak size of the outbreak now?

```
I_peak = out2$I %>%  
  max()
```

```
I_peak
```

```
## [1] 0.6150934
```

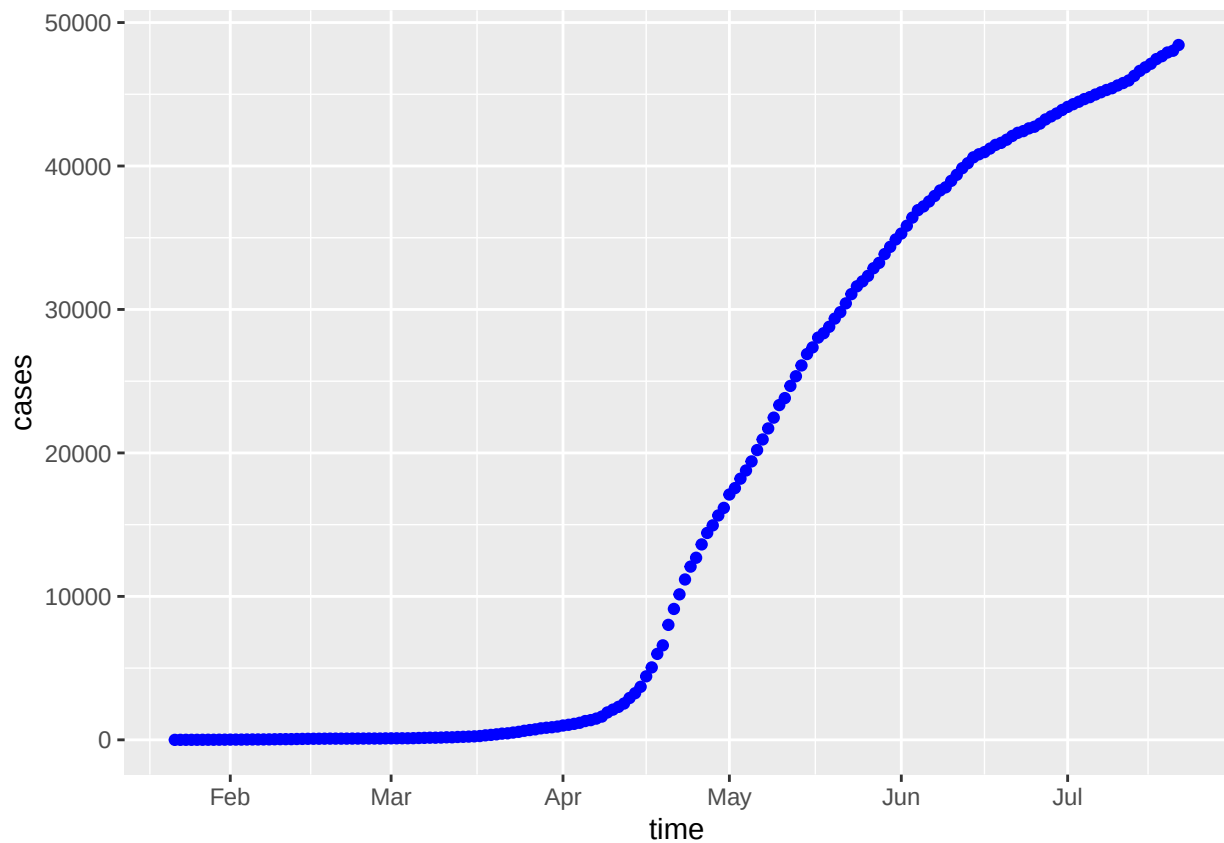
At peak, 61.51% of the population is infected.

2 download the COVID19 data file from canvas (“cases_data_global”). We are going to estimate the early pandemic COVID19 R0 using data from singapore.

```
covid <- load('./cases_data_global.RData')
```

2a) Plot the singapore time series of cases against date.

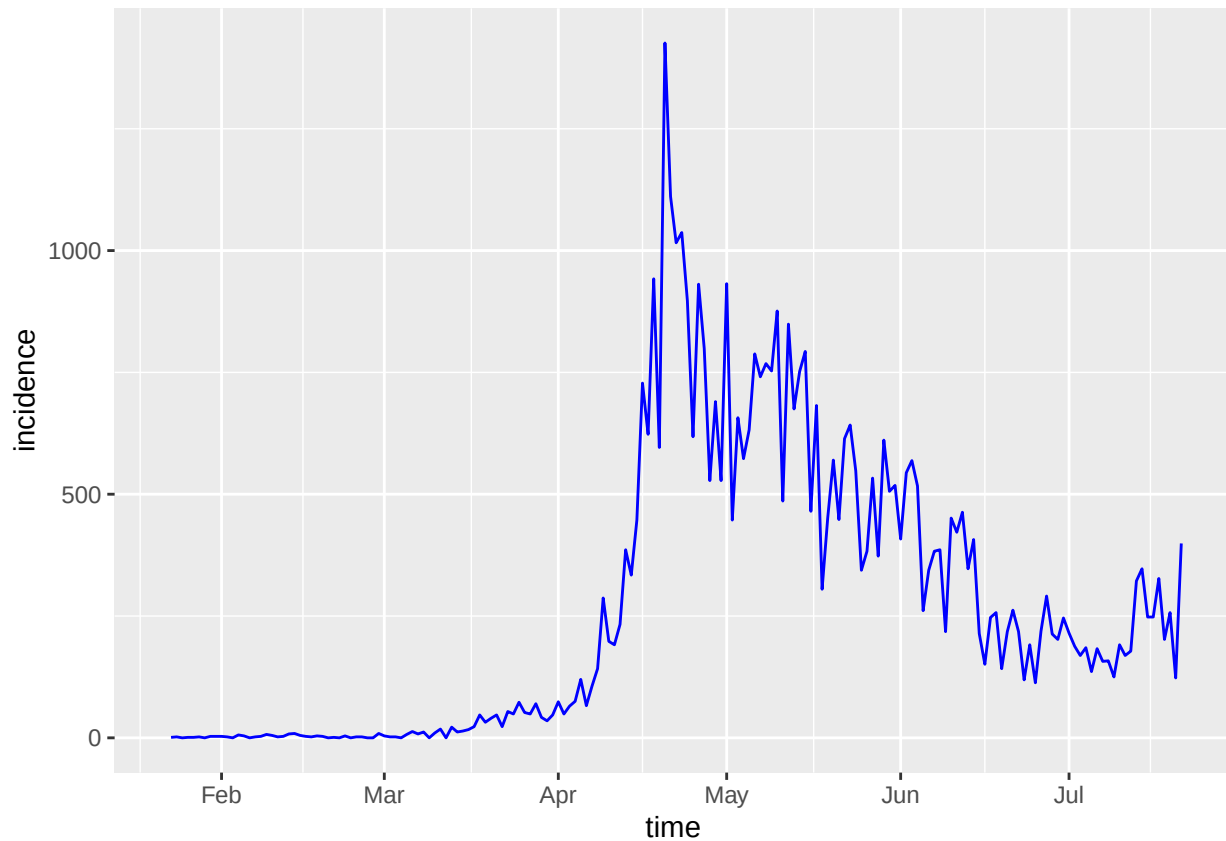
```
sing <- data %>% filter(Country.Region == 'Singapore')  
ggplot(sing, aes(x = time, y = cases)) +  
  geom_point(color = 'blue')
```



2b) The case variable is cumulative. Convert the singapore time series to daily incidence. Plot the incidence.

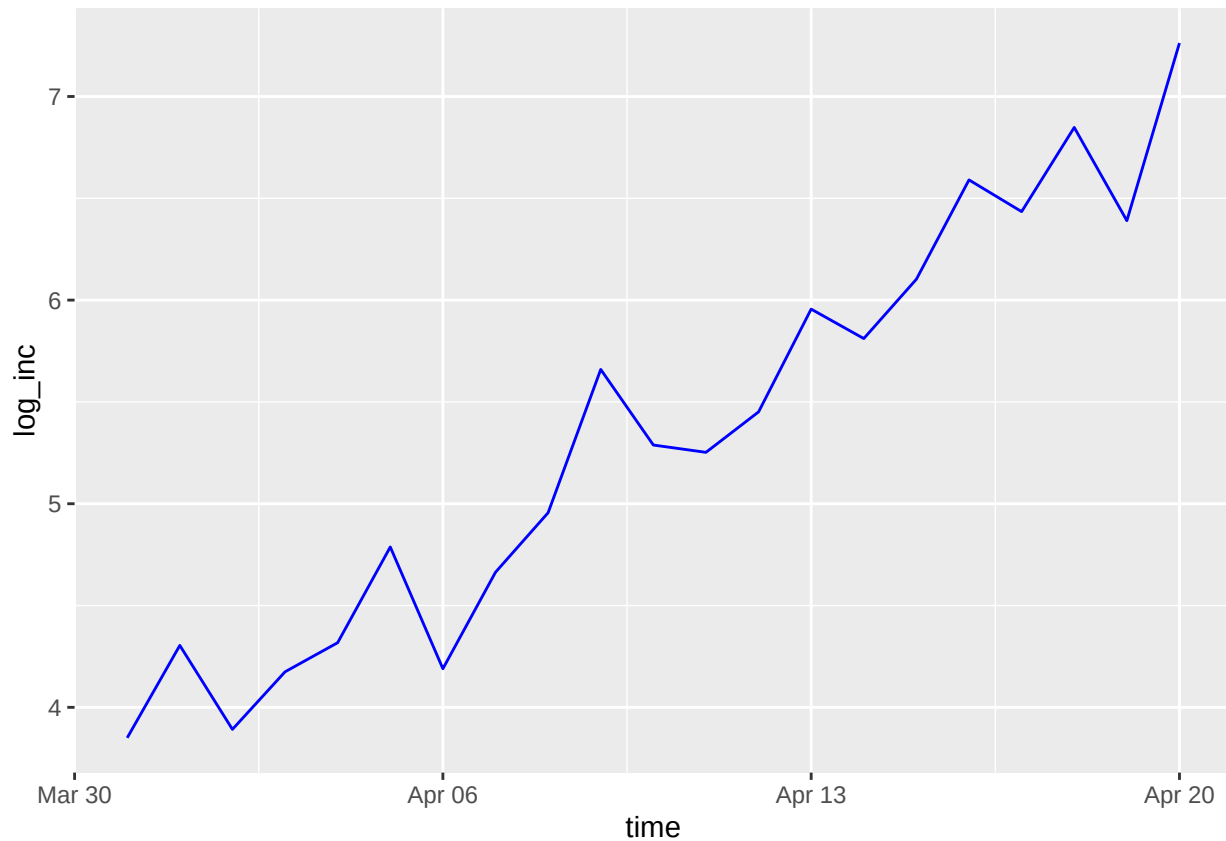
```
sing <- sing %>%  
  mutate(incidence = sing$cases - lag(sing$cases))  
  
ggplot(sing, aes(x = time, y = incidence)) +  
  geom_line(color = 'blue')
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## (`geom_line()`).
```



2c) Look at data from days 70 to 90 when epidemic appears to be growing exponentially. Plot logged daily incidence data between days 70:90 - what do you see? interpret the plot .

```
sing <- sing %>%  
  mutate(log_inc = log(sing$incidence))  
  
ggplot(sing[70:90, ], aes(x = time, y = log_inc)) +  
  geom_line(color = 'blue')
```



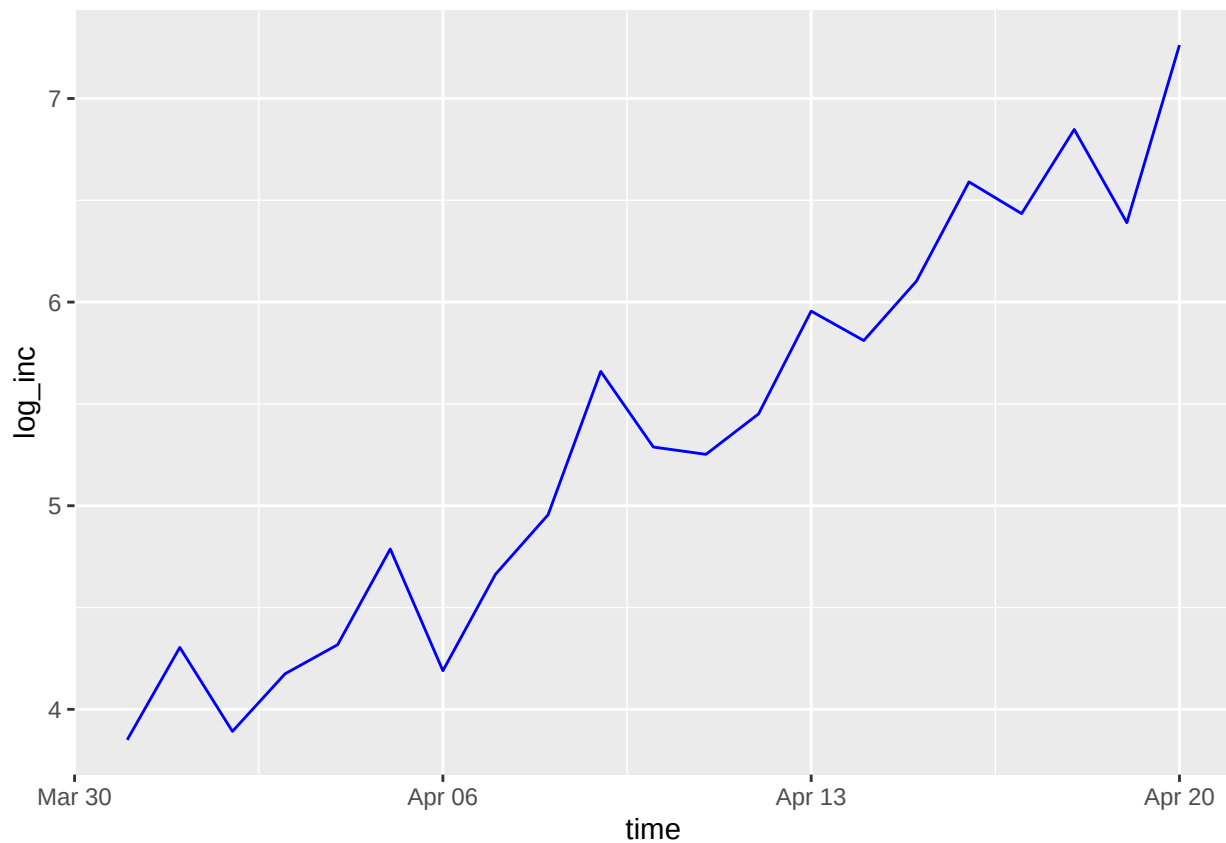
2d) Fit a straight line to the logged incidence data against time. Hint: you'll need to fit a regression line using the `lm` command. Write down the coefficient on time from your regression model.

```
### Remove Inf values
sing_filt <- sing %>%
  filter(is.finite(log_inc))

#filter the filtered values to the same date range as days 70-90
day70 <- sing %>% slice(70) %>% pull(time)
day90 <- sing %>% slice(90) %>% pull(time)

sing_filt <- sing_filt %>%
  filter(time >= as.Date(day70) & time <= as.Date(day90))

ggplot(sing_filt, aes(x = time, y = log_inc)) +
  geom_line(color = 'blue')
```

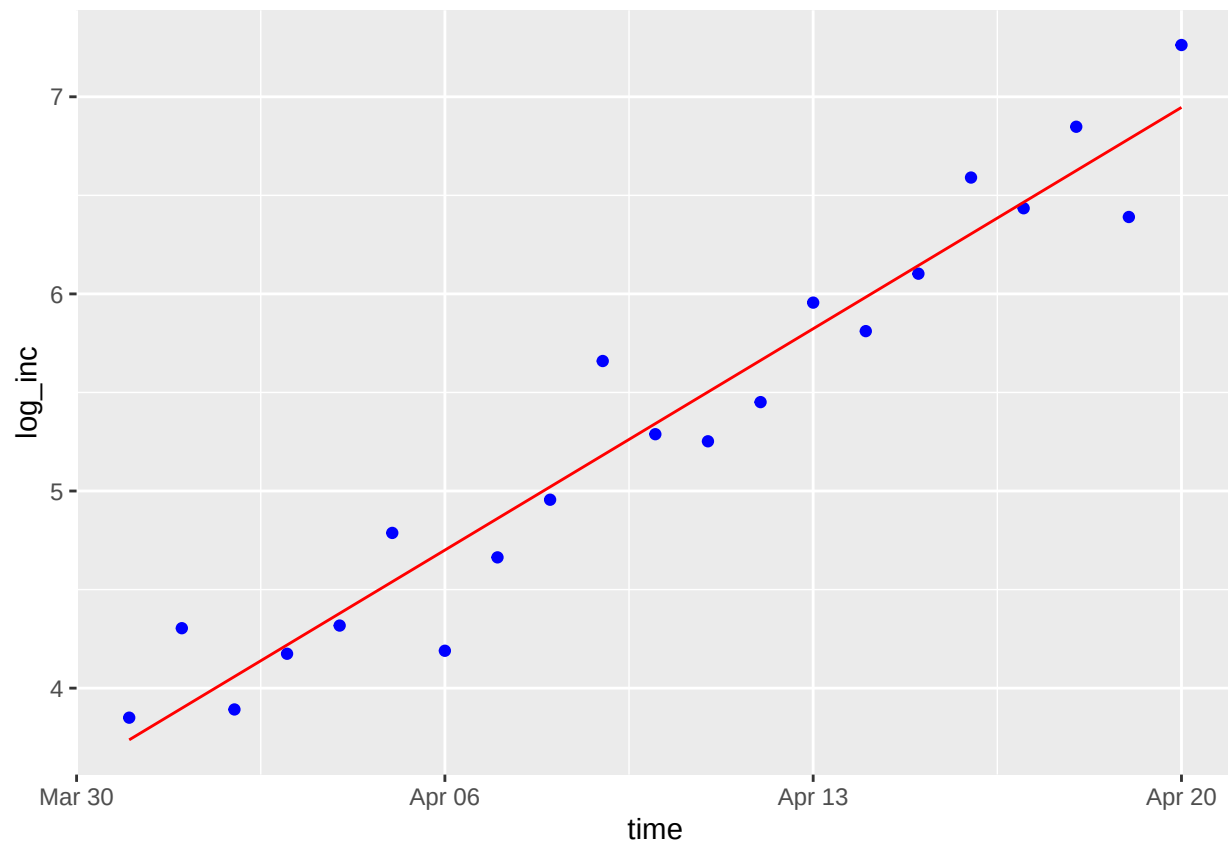
```
model_fit <- lm(log_inc ~ time, data = sing_filt)
model_fit
```

```
##
## Call:
## lm(formula = log_inc ~ time, data = sing_filt)
##
## Coefficients:
## (Intercept)      time
## -2940.4368      0.1604
```

The slope of the line is .1604.

```
time_vec <- sing_filt$time
sing_filt$predicted <- predict(model_fit, newdata = sing_filt)
```

```
ggplot(sing_filt, aes(x = time)) +
  geom_point(aes(y = log_inc), color = 'blue') +
  geom_line(aes(y = predicted), color = 'red')
```



2e) Use the coefficient on time, to estimate R_0 . Assume the infectious period is 10 days.

$R_0 = kD + 1$. $D = 10$ and k is the slope of the line above.

```
k <- coef(model_fit)['time']
```

```
D <- 10
```

```
R0_sing <- k*D + 1
```

```
R0_sing
```

```
## time
```

```
## 2.60428
```

The estimated R_0 is 2.604.